

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of examining, cleaning, summarizing, and visualizing a dataset before applying statistical models or machine learning. Its purpose is to understand the data, discover patterns, identify anomalies, check assumptions, and prepare the data for further analysis. **Main steps in EDA:**

1. Understand the dataset (rows, columns, data types).
2. Clean the data (handle missing values, duplicates, and outliers).
3. Generate summary statistics (mean, median, standard deviation, etc.).
4. Visualize the data using histograms, box plots, scatter plots, and bar charts.
5. Identify relationships and trends among variables.

Benefits:

- Improves data quality.
- Helps choose suitable models.
- Detects errors and unusual observations.
- Provides insights for better decision-making.